Complete Guide to Generated Figures, Graphs, and Tables

Companion Reference for Paper 1 (14-Page Journal Manuscript):

"Who Backs Into the Bay? Benchmarking Information, Incentives, and Sim-to-3D Robustness in Narrow-Corridor Robot Negotiation"

Folder: `papers/paper1_journal/` | Manuscript: `main.pdf` (14 pages) | Generator: `make_figures.py`

Scope of This Document. This guide catalogs every visual artifact and quantitative summary table generated for the 14-page IEEE journal manuscript (`papers/paper1_journal/main.pdf`). Specifically, it documents:

- **7 multi-panel vector PDF figures comprising 20 individual sub-panels, graphs, and schematics** (`fig/*.pdf`), deterministically generated from raw per-episode CSVs and scenario configurations by `make_figures.py`.
- **5 manuscript tables** (`Table I`–`Table V` in `papers/paper1_journal/main.tex`), spanning policy taxonomy, the 18-study 2D + Gazebo experimental design (50,374 2D episodes + 496 Gazebo runs), the 2^3 extreme-loss safeguard ablation, the seed-paired sim-to-3D transfer summary, and the practitioner selection matrix.

For every figure, individual sub-panel, and table, this guide explicitly states **(1) What It Shows** (axes, variables, baselines, sample sizes, uncertainty intervals, and visual encoding) and **(2) What It Means** (physical, algorithmic, or game-theoretic interpretation and key quantitative conclusions).

1 Master Inventory of Generated Figures and Tables

Table 1: Index of all 7 vector PDF figures (20 sub-panels) generated by `papers/paper1_journal/make_figures.py`.

Fig.	PDF File	Generator	Panels	Graphs / Sub-Panels Included	Sec. / Page
Fig. 1	`fig/scenarios.pdf`	`fig1_scenarios`	6	(a) Single-bay map; (b) Two-bay map; (c) Convoy 2v1 map; (d) T-junction map; (e) Gazebo Harmonic 3D Waffle frame; (f) Waffle clearance cross-section	Sec. III-A, p. 4
Fig. 2	`fig/protocol.pdf`	`fig2_protocol`	2	(a) 20 Hz 5-mode agent state machine & safeguards; (b) Snapshot exchange & constant-speed kinematic yield timeline	Sec. III-C, p. 4
Fig. 3	`fig/rq1_information.pdf`	`fig3_rq1_information`	4	(a) Single-bay geometry & FCFS reservation (`S1`); (b) Asymmetric urgency/load/speed (`S2`); (c) Log-normal bid-noise sweep (`S7`); (d) Tabular bandit ID vs. OOD layout shift	Sec. VI, p. 9
Fig. 4	`fig/rq2_incentives.pdf`	`fig4_rq2_incentives`	2	(a) Misreport utility gain curves across payment rules (`S9`); (b) Repeated-encounter delay vs. Jain fairness Pareto (`S10`)	Sec. VII, p. 10
Fig. 5	`fig/rq3_comms_maps.pdf`	`fig5_rq3_comms_maps`	2	(a) Packet-loss & latency scaling up to 99% (`S3+S8b`); (b) Online bay discovery & map-sharing excess (`explore`)	Sec. VIII, p. 11
Fig. 6	`fig/rq4_topologies.pdf`	`fig6_rq4_topologies`	2	(a) Convoy 2-vs.-1 weighted delay with `make_room`; (b) 3-robot T-junction completion & delay before/after `goal_wait`	Sec. IX-A, p. 12
Fig. 7	`fig/rq4_gazebo.pdf`	`fig7_rq4_gazebo`	2	(a) 460-run seed-paired Gazebo-vs.-2D delay scatter; (b) Seed-3 spacetime trace $x(t)$ showing Gazebo lidar-clustering lag	Sec. IX-B, p. 13

Total across Figs. 1–7: $6+2+4+2+2+2+2=20$ individual panels (8 geometric maps/frames/diagrams + 12 quantitative empirical plots).

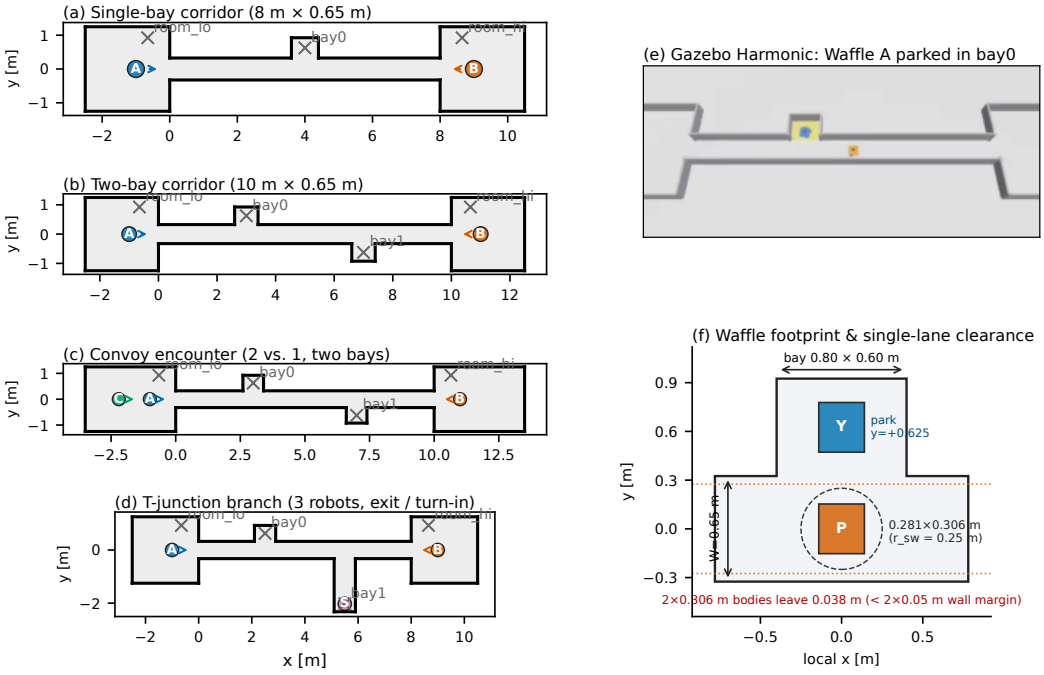
Table 2: Index of all 5 manuscript tables authored in `papers/paper1_journal/main.tex`.

Table	LaTeX Label	Size	Summary Scope & Underlying Source Data	Sec. / Page
Table I	`tab:taxonomy`	9×7	Architectural taxonomy of all 9 policies (`R, S, F, D, P, L, C, G, V`) across comms, inputs, motion model, and incentives (`policy.py`, `planner.py`, `agent.py`)	Sec. IV, p. 6
Table II	`tab:design`	19×6	Complete factorial experimental matrix covering 18 2D studies (50,374 episodes) + ROS 2 / Gazebo validation ($460+36$ runs) grouped by RQ1–RQ4	Sec. V-B, p. 8
Table III	`tab:abl`	8×7	Condensed 2^3 extreme-loss safeguard ablation (`S8c`, 5,760 episodes at 90–99% loss, 2–8 s latency) from `S8c_protocol_ablation.csv.gz`	Sec. VIII-A, p. 10
Table IV	`tab:gz`	16×6	Seed-paired ROS 2 Jazzy / Gazebo Harmonic transfer results across 14 main scenario groups (460 runs) + 2 Batch-2 groups (36 runs)	Sec. IX-B, p. 11
Table V	`tab:guide`	5×4	Practitioner policy and protocol selection matrix mapping 5 warehouse deployment regimes to recommended policies, flags, and quantitative trade-offs	Sec. X-A, p. 12

2 Detailed Guide to Generated Figures and Sub-Panel Graphs

2.1 Figure 1: Physical Problem Setup, Topologies, and Clearance Geometry

File / Label: `fig/scenarios.pdf (label: fig:scenarios)` | **Manuscript Location:** Section III-A, Page 4
Generator / Source: `make_figures.py :: fig1_scenarios()`
Primary Data Inputs: `configs/scenarios/corridor_single_bay.yaml`, `corridor_two_bays.yaml`, `corridor_convoy.yaml`, `corridor_tjunction_goal.yaml`, and `media/gazebo_final/corridor_cost_bid.webp`

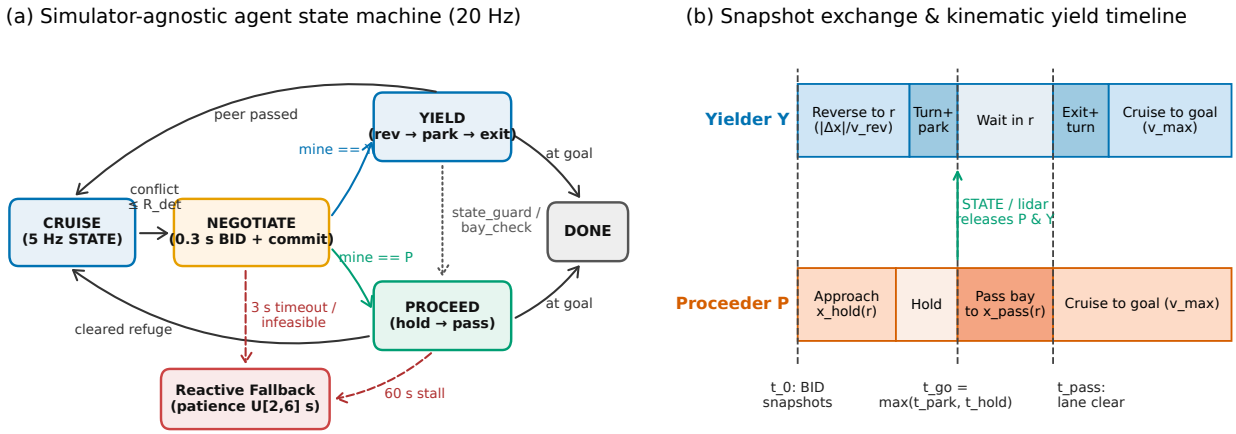


Panel-by-Panel Breakdown:

- **Panels (a)–(d) — Benchmark Corridor Topologies:**
  - *What They Show:* Four 2D metric floorplans (x vs. y in meters) rendered directly from the scenario YAML files used by both simulators, showing corridor walls, Staging Rooms (`room_lo` at $x < 0$, `room_hi` at $x > L$), lateral pull-in refuge bays (0.80×0.60 m), and initial robot poses/headings: (a) single-bay straight corridor (8.0×0.65 m) with opposing robots A (blue) and B (orange); (b) two-bay corridor (10.0×0.65 m) with opposing bays at $x = 3$ m and $x = 7$ m; (c) 2-vs.-1 convoy encounter where follower C (green) trails leader A against oncoming B; and (d) 3-robot T-junction where a 2.0 m deep side branch at $x = 7$ m (`bay1`) hosts robot S (purple) while A and B contest the main aisle and upstream `bay0` ($x = 3$ m).
  - *What They Mean:* Establish the progressive topological complexity of the benchmark—moving from an isolated binary choice of *who* yields in a single-bay aisle (a), to *where* and *with what map knowledge* to yield in multi-bay aisles (b), to coupled multi-robot chain reactions where pairwise negotiation alone deadlocks without follower `make_room` (c) or upstream `goal_wait` (d).
- **Panel (e) — ROS 2 Jazzy / Gazebo Harmonic 3D Execution Frame:**
  - *What It Shows:* An overhead snapshot extracted from an actual Gazebo Harmonic run (frame 48 of `corridor_cost_bid.webp`), depicting TurtleBot3 Waffle A parked off-lane inside `bay0` while Waffle B passes unimpeded along the $y = 0$ corridor centerline.
  - *What It Means:* Demonstrates visually that the exact same 20 Hz negotiation state machine executes on full 3D rigid-body differential-drive models with simulated 360° GPU ray-cast lidar and extruded wall geometry.
- **Panel (f) — Waffle Chassis Footprint and Single-Lane Clearance Cross-Section:**
  - *What It Shows:* Exact metric cross-section of the $W = 0.65$ m aisle and 0.80×0.60 m pull-in bay, plotting the rectangular TurtleBot3 Waffle chassis (0.281×0.306 m), its $R_{sw} = 0.25$ m turn-in-place swept circle, the 0.05 m safety wall margins (orange dotted lines), Proceeder P centered at $y = 0$, and Yielder Y parked at $y_{park} = +0.625$ m.
  - *What It Means:* Proves geometrically *why continuous local collision avoidance (e.g., ORCA/CBF) cannot solve this problem without discrete bay negotiation*: two 0.306 m-wide Waffle bodies side-by-side consume 0.612 m inside a 0.650 m corridor, leaving only 0.038 m total clearance—violating the required $2 \times 0.05 = 0.10$ m wall margin and making lateral passing physically impossible anywhere outside a refuge bay.

2.2 Figure 2: Simulator-Agnostic Protocol State Machine and Kinematic Yield Timeline

File / Label: `fig/protocol.pdf (label: fig:proto)` | **Manuscript Location:** Sections III-C & IV-B, Page 4
Generator / Source: `make_figures.py :: fig2_protocol()`
Primary Data Inputs: Architectural specification of `amr_negotiation/agent.py` and `amr_negotiation/planner.py` (Eqs. 1–2)

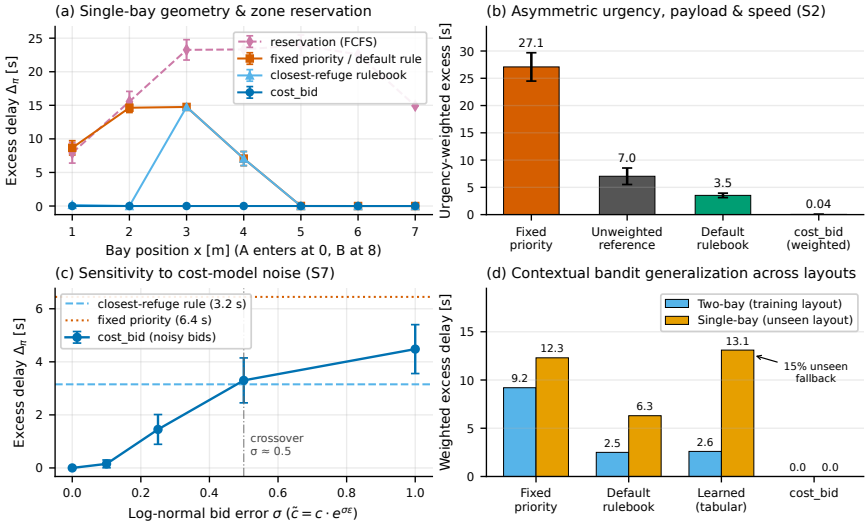


Panel-by-Panel Breakdown:

- **Panel (a) — Simulator-Agnostic 20 Hz Agent State Machine:**
  - *What It Shows:* A directed finite-state machine diagram depicting the five primary control modes (`CRUISE` with 5 Hz `STATE` beacons, `NEGOTIATE` with 0.3 s `BID` broadcasts and optional `commit` confirmation, `YIELD` executing reverse $\rightarrow$ turn $\rightarrow$ park $\rightarrow$ exit, `PROCEED` executing approach $\rightarrow$ hold $\rightarrow$ pass, and `DONE`), alongside the `Reactive Fallback` mode (random patience $U[2,6]$ s) triggered on 3 s negotiation timeout, infeasible refuge sets, or 60 s stall timeouts, plus the cross-transitions for `state_guard` and `bay_check`.
  - *What It Means:* Clarifies the strict architectural separation between **Layer 2 (Protocol & Execution)**, which handles asynchronous message drops, timeouts, and physical safety guards identically across all policies, and **Layer 1 (Decision Mechanism)**, which plugs into `NEGOTIATE` purely to map frozen snapshots (S_A, S_B) to (Y, r). It also makes explicit the exact safeguard edges later ablated in Table III (`S8c`).
- **Panel (b) — Snapshot Exchange and Constant-Speed Kinematic Yield Timeline:**
  - *What It Shows:* A synchronized two-track horizontal timeline comparing Yielder Y's five motion phases (reverse $|x_r - x_Y|/v_{\text{rev}}$, 90° turn + park into y_{park}, wait in refuge r, exit + turn, and cruise at $v_{\max}$) against Proceeder P's four motion phases (approach $x_{\text{hold}}(r)$, hold if Y is not yet parked, pass bay to $x_{\text{pass}}(r)$, and cruise to goal), anchored by synchronization instants t_0 (`BID` snapshot freeze), $t_{\text{go}} = \max(t_{\text{park}}, t_{\text{hold}})$, and t_{pass} (lane clear release via `STATE` or lidar).
  - *What It Means:* Visually explains Equations (1)–(2) of the paper: why yielding delay is non-linear in bay position (reversing uses slower $v_{\text{rev}} = 0.18$ m/s plus two 90° turns and lateral bay entry/exit, while P only waits $\max(0, t_{\text{park}} - t_{\text{hold}})$), and how `cost_bid` and `clarke` price candidate outcomes in sub-millisecond CPU time without forward physics rollouts.

2.3 Figure 3: RQ1 — Information Fidelity, Geometry, Bid Noise, and Layout Shift

File / Label: `fig/rq1_information.pdf` (`label: fig:rq1`) | **Manuscript Location:** Section VI (RQ1), Page 9
Generator / Source: `make_figures.py :: fig3_rq1_information()`
Primary Data Inputs: `data/raw/classical/S1_baseline.csv.gz`, `reservation.csv.gz`, `S2_asymmetry.csv.gz`, `S7_noisy_bids.csv.gz`, and `data/tables/classical/learned_negotiator.md`

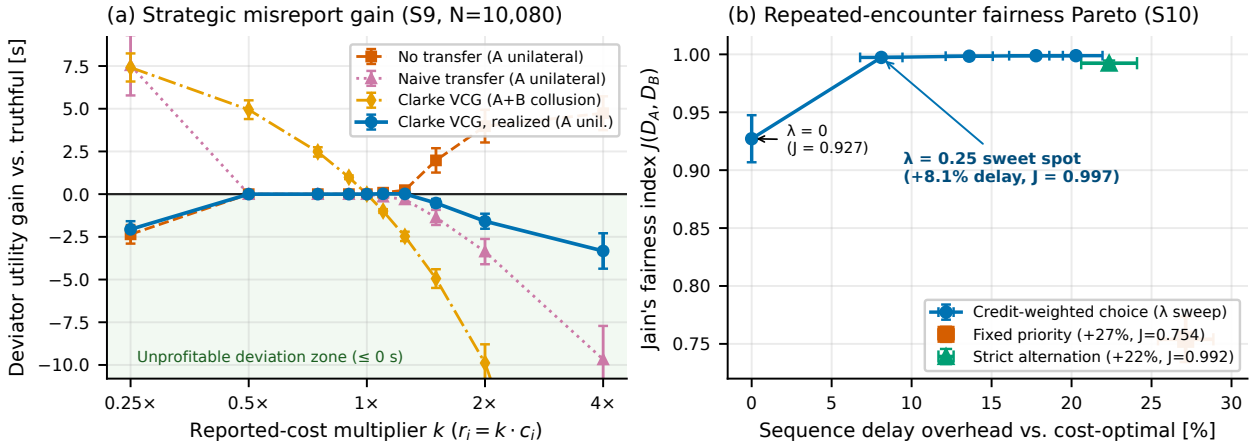


Panel-by-Panel Breakdown:

- **Panel (a) — Single-Bay Geometry Sweep and Zone Reservation** (`S1` + `reservation`)**:**
  - *What It Shows:* Mean paired excess delay $\Delta_\pi = D_\pi - \min(D_A, D_B)$ [s] ($\pm$95% CI) on the y-axis plotted against single-bay position $x \in \{1,2,3,4,5,6,7\}$ m on the x-axis (A enters at $x = 0$, delayed B at $x = 8$ m), comparing FCFS corridor `reservation` ($N = 1{,}260$, purple diamonds), `fixed_priority/rules_default` ($N = 2{,}940$, orange squares), `rules_proximity` (sky-blue triangles), and `cost_bid` (blue circles).
  - *What It Means:* Demonstrates three core RQ1 findings: (1) industrial FCFS corridor locking (`reservation`) wastes 18.8 ± 0.9 s excess delay across all bay positions because holding outside the aisle requires turning off-lane in `room_hi` and forfeits mid-corridor bays entirely; (2) static ID priority and seniority-first rulebooks suffer severe 14.6–14.8 s spikes at $x = 2$–3 m by forcing B to reverse nearly the whole corridor even when A just passed a bay; and (3) promoting geometric proximity (`rules_proximity`) halves excess delay to 3.1 s, while kinematic `cost_bid` achieves $\Delta_\pi = 0.0$ s at every bay coordinate (100% hindsight-optimal yielder match).
- **Panel (b) — Heterogeneous Robot Urgency, Payload, and Speed** (`S2`, $N = 5{,}040$)**:**
  - *What It Shows:* Bar chart with 95% CI error bars comparing urgency-weighted excess delay [s] against the weighted hindsight reference (`oracle_w`) across four policies on asymmetric encounters ($u_B \in \{1,3\}$, payload $\{0,1\}$, $v_B \in \{0.26, 0.13\}$ m/s): `fixed_priority` (27.1 s), unweighted hindsight `oracle` (7.0 s), `rules_default` (3.5 s), and weighted `cost_bid` (0.04 s).
  - *What It Means:* Proves that *objective alignment matters more than kinematic precision*: even an omniscient oracle optimizing unweighted time loses 7.0 s of urgency-weighted delay—twice the 3.5 s excess of a simple 6-rule etiquette (`rules_default`) that respects urgency and payload. Weighted `cost_bid` captures virtually the entire remaining gap (0.04 ± 0.02 s, matching `oracle_w` on $704/720 = 98\%$ of encounters).
- **Panel (c) — Sensitivity to Multiplicative Bid-Estimation Noise** (`S7`, $N = 3{,}150$)**:**
  - *What It Shows:* Excess delay Δ_π [s] ($\pm$95% CI) of `cost_bid` plotted against log-normal multiplicative bid noise standard deviation $\sigma \in \{0, 0.1, 0.25, 0.5, 1.0\}$ ($\tilde{c} = c \cdot e^{\sigma\varepsilon}$), benchmarked against horizontal reference lines for noise-free `rules_proximity` (3.2 s) and `fixed_priority` (6.4 s), with the crossover marked at $\sigma \approx 0.5$.
  - *What It Means:* Pinpoints the exact model-fidelity threshold required for auction-based coordination: `cost_bid` strictly dominates heuristic rulebooks when travel-time estimation error is $\sigma \leq 0.25$ ($\Delta_\pi \leq 1.5$ s), reaches statistical parity with the closest-refuge rulebook at $\sigma = 0.5$ ($\sim 1.65\times$ multiplicative error, 3.3 s vs. 3.2 s), and degrades below the rulebook for $\sigma > 0.5$ while still beating static ID priority even at $\sigma = 1.0$ (4.5 s vs. 6.4 s).
- **Panel (d) — Tabular Contextual Bandit Generalization Across Corridor Layouts** (`learned`)**:**
  - *What It Shows:* Grouped bar chart comparing weighted excess delay [s] on the held-out *training* two-bay layout (light blue bars, $n = 150$) vs. an *unseen* single-bay layout with random bay coordinates (gold bars, $n = 150$) across `fixed_priority`, `rules_default`, `learned` (trained offline on 800 forced two-bay pairs, 93 populated state keys), and `cost_bid`.
  - *What It Means:* Shows that while a model-free tabular bandit easily matches the engineered rulebook in-distribution (+2.6 s vs. +2.5 s), it suffers severe out-of-distribution failure under layout shift (+13.1 s, worse than fixed priority's +12.3 s, with 15% unseen-key fallbacks). Coarse ego-centric feature bins memorize layout-specific bay priors rather than transferable stopping-distance physics.

2.4 Figure 4: RQ2 — Strategic Misreporting Incentives and Repeated Fairness

File / Label: `fig/rq2_incentives.pdf (label: fig:rq2)` | **Manuscript Location:** Section VII (RQ2), Page 10
Generator / Source: `make_figures.py :: fig4_rq2_incentives()`
Primary Data Inputs: `data/raw/classical/S9_payments.csv.gz` (10,080 eps) and `data/raw/classical/S10_repeated.csv.gz` (50×20 encounters)

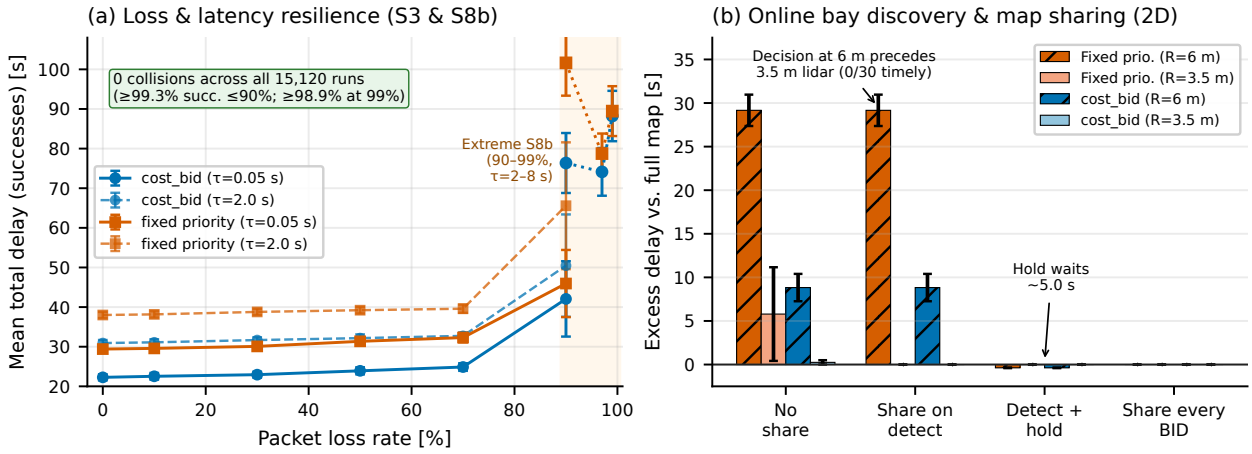


Panel-by-Panel Breakdown:

- **Panel (a) — Single-Encounter Strategic Misreport Gain Across Payment Mechanisms** (S9, $N = 10{,}080$)**:**
  - *What It Shows:* Mean deviator quasi-linear utility gain [s] relative to truthful bidding ($k = 1\times$) ($\pm 95\%$ CI) on the y-axis vs. logarithmic report-scaling multiplier $k \in [0.25\times, 4\times]$ ($r_i = k \cdot c_i$) on the x-axis. The green shaded region (≤ 0 s) denotes unprofitable lies. Four mechanisms/coalitions are plotted: unpriced `cost_bid` unilateral A (orange dashed squares), naive "proceeder-pays-yielder's-bid" transfer unilateral A (purple dotted triangles), joint $A+B$ collusion under Clarke VCG (gold dash-dotted diamonds), and realized-trajectory unilateral A under Clarke VCG (solid blue circles).
  - *What It Means:* Proves empirically both the power and boundaries of VCG mechanism design in physical robotics: (1) unpriced `cost_bid` rewards bid inflation ($+4.74$ s gain at $k = 4\times$ by forcing the peer to yield); (2) naive direct transfers backfire by rewarding bid *under-reporting* ($+7.54$ s gain at $k = 0.25\times$); (3) Clarke pivot payments completely eliminate unilateral misreport gains (0.00 s predicted; $\leq +0.02 \pm 0.02$ s on realized simulator trajectories, bending downward to -3.3 s at extreme k); yet (4) joint $A+B$ bid shading ($k < 1$) beats the external bank ($+4.94$ s combined gain at $0.5\times$, $+7.42$ s at $0.25\times$) because scaling both bids down preserves the allocation while shrinking both pivot charges.
- **Panel (b) — Repeated-Encounter Efficiency vs. Cumulative Fairness Pareto Frontier** (S10)**:**
  - *What It Shows:* A 2D trade-off scatter/curve over 50 random 20-encounter sequences (with horizontal and vertical 95% CI bars), plotting percentage sequence delay overhead above pure cost-optimal selection ($\lambda = 0$) on the x-axis against Jain's cumulative fairness index $J(D_A^{\text{cum}}, D_B^{\text{cum}}) \in [0.5, 1.0]$ on the y-axis. Compares the cumulative-delay credit sweep $\lambda \in \{0, 0.25, 0.5, 1.0, 2.0\}$ (Eq. 6, blue curve) against `fixed_priority` (orange square) and strict turn-taking `alternate` (green triangle).
  - *What It Means:* Shows that in cooperative single-owner fleets, VCG transfers are unnecessary to achieve long-run equity: static `fixed_priority` is both unfair ($J = 0.754$) and slow ($+27.1\%$ overhead), while strict alternation reaches $J = 0.992$ only by wasting $+22.3\%$ delay when bays are asymmetric. A lightweight non-monetary fairness weight of $\lambda = 0.25$ hits a sharp Pareto sweet spot—achieving near-perfect cumulative equity ($J = 0.997$) for just $+8.1\%$ delay overhead by flipping only close-call encounters toward the historically burdened robot.

2.5 Figure 5: RQ3 — Channel Impairment Resilience and Online Map Sharing

File / Label: `fig/rq3_comms_maps.pdf (label: fig:rq3)` | **Manuscript Location:** Section VIII (RQ3), Page 11
Generator / Source: `make_figures.py :: fig5_rq3_comms_maps()`
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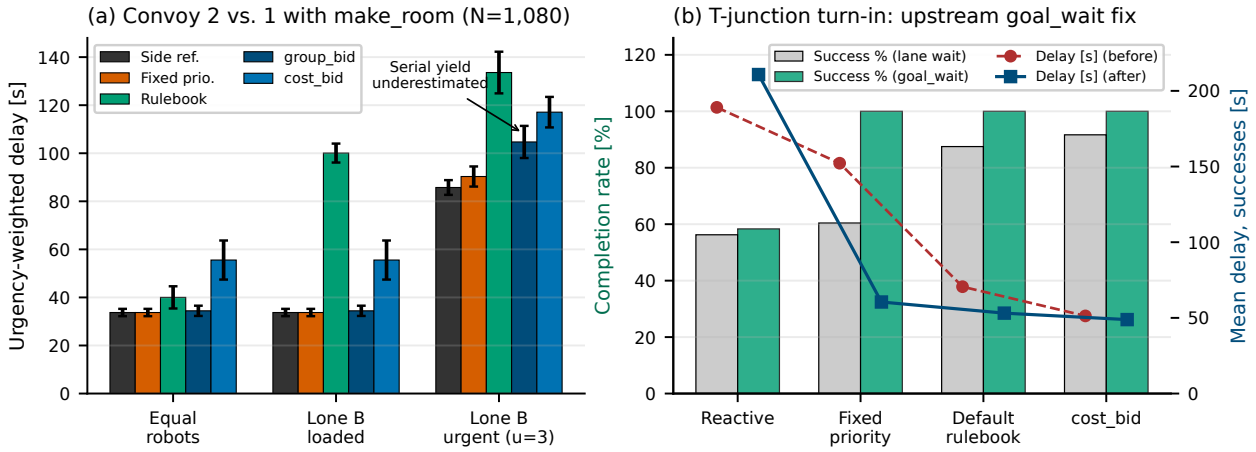


Panel-by-Panel Breakdown:

- **Panel (a) — Packet Loss and Latency Scaling Up to 99% Loss (`S3` + Extreme `S8b`):**
  - *What It Shows:* Mean total delay over successful episodes [s] (±95% CI) vs. Bernoulli packet drop rate from 0% to 99%, plotting `cost_bid` (blue circles) and `fixed_priority` (orange squares) at base latencies $\tau = 0.05$ s (solid) and $\tau = 2.0$ s (dashed) from `S3` ($v_{\max} = 0.26$ m/s, 3,600 eps), seamlessly joined on the right (amber shaded band, 90–99% loss) by extreme stress study `S8b` ($\tau = 2$–8 s, $v_{\max} = 0.50$ m/s, 1,440 eps).
  - *What It Means:* Demonstrates remarkable protocol grace under harsh industrial Wi-Fi degradation: across all 15,120 communication and ablation runs (`S3+S8+S8b+S8c`), **zero collisions occurred**, with completion $\geq 99.3\%$ up to 90% loss and 98.9–99.2% at 99% loss. Periodic 0.3 s `BID` broadcasts keep delay completely flat (< 2 s rise) up to 50% packet loss, and `cost_bid` preserves its ~ 8–10 s delay advantage over `fixed_priority` across every loss and latency tier.
- **Panel (b) — Online Bay Discovery and Map-Sharing Modes in 2D (`explore`, 0% Loss):**
  - *What It Shows:* Grouped bar chart (±95% CI) of paired excess delay [s] relative to full prior map knowledge across four map-sharing configurations—`No share`, `Share on detect`, `Detect + hold` (`hold_for_map`), and `Share every BID`—comparing `fixed_priority` and `cost_bid` at conflict detection ranges $R_{\text{det}} = 6.0$ m (hatched) vs. $R_{\text{det}} = 3.5$ m (solid) (1,800-episode `explore_map` dataset).
  - *What It Means:* Reveals a critical cross-layer timing trap between radio negotiation range and onboard sensor range: at default $R_{\text{det}} = 6.0$ m, robots freeze snapshots and commit to a yield plan *before* their 3.5 m lidars ever detect each other (0/30 timely maps), making `share_on_detect` just as costly as never sharing maps (+29.2 s for `fixed_priority`, +8.8 s for `cost_bid`). While `hold_for_map` or shrinking R_{det} to 3.5 m fixes this in 2D (≈ 0 s excess), unconditionally piggybacking known bay IDs on every `BID` (`Share every BID`) achieves +0.0 s excess immediately at 6.0 m with zero holding delay.

2.6 Figure 6: RQ4a — Multi-Robot Convoys and T-Junction Topologies

File / Label: `fig/rq4_topologies.pdf (label: fig:rq4_topo)` | **Manuscript Location:** Section IX-A (RQ4), Page 12
Generator / Source: `make_figures.py :: fig6_rq4_topologies()`
Primary Data Inputs: `data/raw/classical/convoy_group.csv.gz` (1,080 eps), `tjunction_goal.csv.gz` (384 eps), and `data/tables/classical/tjunction_goal_before.txt`

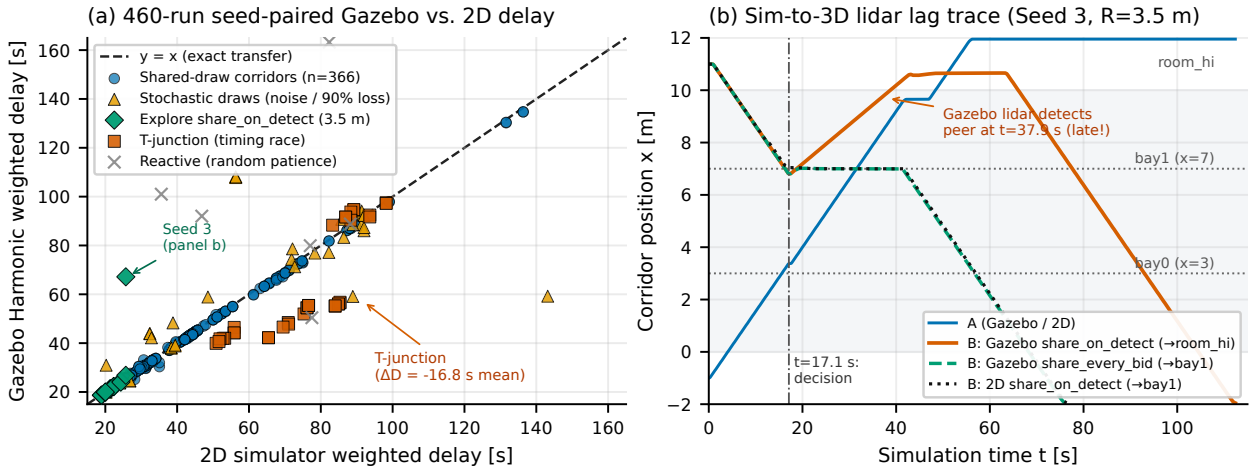


Panel-by-Panel Breakdown:

- **Panel (a) — Two-vs.-One Convoy Encounters with Follower `make_room` ($N = 1{,}080$):**
  - *What It Shows:* Grouped bar chart (±95% CI) of urgency-weighted delay [s] across three opponent regimes—`Equal robots` ($u = 1$), `Lone B loaded`, and `Lone B urgent` ($u_B = 3$)—comparing hindsight `side_oracle` (charcoal), `fixed_priority` (orange), `rules_default` (green), side-level `group_bid` (dark blue), and pairwise `cost_bid` (medium blue), with follower `make_room` enabled (72/72 completion in every cell).
  - *What It Means:* Shows both the necessity and the modeling subtlety of group-level bidding: at equal urgency, pairwise `cost_bid` prices only leader A and ignores follower C (55.6 s), whereas `group_bid` prices the whole convoy and slashes delay to 34.4 s (matching `side_oracle`'s 33.7 s). However, when lone B has $u_B = 3$, `group_bid` (104.7 s) beats pairwise `cost_bid` (117.1 s) yet trails `fixed_priority` (90.3 s) and `side_oracle` (85.7 s) because Eq. (4) sums member delays additively rather than modeling the *serial* wait while follower C backs out of leader A's reversal path.
- **Panel (b) — Three-Robot T-Junction Turn-In Before vs. After Upstream `goal_wait`:**
  - *What It Shows:* Dual-axis chart (0% loss, $n = 48$ episodes per policy) comparing episode completion rate [%] on the left axis (gray bars = before fix, holding in main lane; green bars = after adding upstream bay staging `goal_wait`) and mean successful-episode delay [s] on the right axis (red dashed circles = before; dark-blue solid squares = after) across `reactive`, `fixed_priority`, `rules_default`, and `cost_bid`.
  - *What It Means:* Demonstrates that three-robot branch junctions induce coupled gridlock when a robot whose branch goal (`bay1`) is occupied waits in the main stem while an oncoming robot approaches. Adding the topological `goal_wait` rule (parking temporarily in upstream `bay0` until `bay1` clears) lifts negotiating-policy completion from 60.4%–91.7% to **100%** (48/48) across all three negotiating policies while cutting `fixed_priority` delay from 152.2 s to 60.5 s and achieving 48.9 ± 4.0 s under `cost_bid`.

2.7 Figure 7: RQ4b — 496-Run ROS 2 / Gazebo Transfer and Lidar-Lag Trace

File / Label: `fig/rq4_gazebo.pdf (label: fig:rq4_gz)` | **Manuscript Location:** Section IX-B (RQ4), Page 13
Generator / Source: `make_figures.py :: fig7_rq4_gazebo()`
Primary Data Inputs: `data/tables/gazebo/gazebo_main_runs.csv`, `gazebo_all_runs.csv`, `media/gazebo_final/explore_det35_fixed_priority.json`, `explore_all35_fixed_priority.json`, and `sim2d`



Panel-by-Panel Breakdown:

- **Panel (a) — 460-Run Seed-Paired Gazebo Harmonic vs. 2D Weighted Delay Scatter:**
  - *What It Shows:* Per-episode scatter plot of Gazebo urgency-weighted delay [s] (y-axis) against identical-seed 2D simulator weighted delay [s] (x-axis) across all paired successful episodes in the 460-run main grid, stratified by regime relative to the $y = x$ exact-transfer diagonal: 366 shared-draw straight/convoy corridor runs (blue circles), independent stochastic draws under bid noise or 90% loss (gold triangles), 3.5 m `explore share_on_detect` runs (green diamonds), 36 T-junction runs (orange squares), and `reactive` random-patience runs (gray crosses).
  - *What It Means:* Visually separates where fast 2D kinematic benchmarking is sufficient from where 3D physics-in-the-loop simulation is mandatory: all 366 shared-draw corridor runs cluster tightly along $y = x$ (policy-level excess delays agree within ≤ 1.0 s and yielder choices match in 96.8% of negotiating runs), whereas systematic off-diagonal clusters pinpoint two genuine sim-to-3D transfer gaps—Gazebo lidar peer-clustering lag (green diamonds above $y = x$, including annotated Seed 3) and three-robot T-junction stem-release timing races (orange squares below $y = x$, $\Delta D = -16.8$ s mean).
- **Panel (b) — Empirical Sim-to-3D Spacetime Trajectory $x(t)$ on Seed 3 ($R_{\text{det}} = 3.5$ m):**
  - *What It Shows:* Longitudinal corridor position $x(t) \in [-2, 12]$ m vs. simulation time $t \in [0, 115]$ s for Seed 3 under `fixed_priority` in `explore` mode ($R_{\text{det}} = 3.5$ m), marking `bay0` ($x = 3$ m), `bay1` ($x = 7$ m), and `room_hi` ($x > 10$ m). Compares Robot A's trajectory (solid blue) against three paired trajectories for yielding Robot B: Gazebo `share_on_detect` (solid orange, reversing all the way to `room_hi`), Gazebo `share_every_bid` (dashed green, ducking into `bay1`), and 2D `share_on_detect` (dotted black, also ducking into `bay1`).
  - *What It Means:* Provides the exact physical mechanism behind the `explore` sim-to-3D discrepancy: in 2D, idealized range check detects peer A inside 3.5 m before the $t = 17.1$ s decision freeze so A shares `bay1` in time; in 3D Gazebo, wall-filtered `LaserScan` clustering (`cluster_dist` 0.35 m) does not register oncoming Waffle A's small front profile until 2.3–2.6 m separation (occurring late at $t = 37.9$ s after B has already reversed to `room_hi`), costing ~ 41 s of unnecessary reversing on Seed 3. Unconditional `share_map` on every `BID` completely eliminates this 3D sensor-clustering bottleneck.

3 Detailed Guide to All 5 Manuscript Tables

3.1 Table I: Taxonomy of Benchmarked Corridor Coordination Policies

File / Label: `Table I (label: tab:taxonomy)` | **Manuscript Location:** Section IV, Page 6
Generator / Source: Authored in `papers/paper1_journal/main.tex` (lines 139–156)
Primary Data Inputs: Synthesizes exact implementation properties of `amr_negotiation/policy.py`, `planner.py`, and `agent.py`

- **What It Shows:** A 9-row × 7-column architectural comparison of every policy evaluated in the paper—`reactive` (R), `reservation` (S), `fixed_priority` (F), `rules_default` (D), `rules_proximity` (P), `learned` (L), `cost_bid` (C), `group_bid` (G), and `clarke` (V)—classified across *Family*, *Comms Required* (None, `STATE` only, or `STATE+BID`), *State Inputs Used*, *Motion Model* (None, Feasibility only, Refuge distance, Offline sim table, Constant-speed, or Side-level sum), *Strategic Incentive Property*, and *Multi-Robot / Map Scope*, with explicit footnotes clarifying non-bid attack surfaces and VCG budget-surplus/collusion caveats.
- **What It Means:** Gives readers a unified conceptual ladder ordered by increasing information and modeling fidelity—showing that all negotiating policies share the exact same radio packet format and execution layer, differing solely in how much physical state, kinematic modeling, and game-theoretic pricing they apply to select (Y, r).

3.2 Table II: Complete Factorial Experimental Design Matrix

File / Label: `Table II (label: tab:design)` | **Manuscript Location:** Section V-B, Page 8
Generator / Source: Authored in `papers/paper1_journal/main.tex` (lines 247–277)
Primary Data Inputs: `experiments/classical/*.py`, `data/raw/classical/*.csv.gz`, and `experiments/gazebo/make_scen_all.py`

- **What It Shows:** A 19-row × 6-column master experimental table organized into four horizontal blocks matching **RQ1** (6 studies: `S1`, `reservation`, `S4`, `S2`, `S7`, `learned`), **RQ2** (3 studies: `S6`, `S9`, `S10`), **RQ3** (7 studies: `S3`, `S8`, `S8b`, `S8c`, Asymmetric maps, Blocked bays, Explore & hold), and **RQ4** (3 studies: Convoy 2v1, 3-robot T-junction, and ROS 2 / Gazebo 3D). For each study, it lists exact factorial grid parameters, random seed counts (2–60), evaluated policy codes, and exact episode counts N (50,374 2D episodes + $460 + 36 = 496$ Gazebo episodes).
- **What It Means:** Establishes full statistical transparency and reproducibility across the entire study suite, making immediately clear which studies share identical episode draws (enabling tight paired excess-delay confidence intervals) and how the 496 Gazebo runs stratify across every 2D study family.

3.3 Table III: Extreme-Loss Factorial Safeguard Ablation

File / Label: `Table III (label: tab:abl)` | **Manuscript Location:** Section VIII-A (RQ3), Page 10
Generator / Source: Authored in `papers/paper1_journal/main.tex` (lines 344–362)
Primary Data Inputs: `data/raw/classical/S8c_protocol_ablation.csv.gz` (5,760 episodes; verified via `data/tables/classical/round2_studies.md`)

- **What It Shows:** Reports the 2^3 full-factorial ablation of protocol safeguards (`commit` $\in \{$on, off$\}$ × `state_guard` $\in \{$on, off$\}$ × `bay_check` $\in \{$on, off$\}$) for `fixed_priority` and `cost_bid` across the worst-case `S8b` channel grid (90–99% packet loss, 2–8 s latency, $v_{\max} = 0.5$ m/s; 360 episodes per row, condensed to 8 distinct rows because `bay_check` had zero triggers and identical episode-by-episode metrics). Columns report *Success %*, *Deadlock count*, *Mean Delay* $\pm$ *95% CI [s]*, and *Transient Dual-Yield %*, noting **0 collisions in every arm** (95% upper bound 1.06%).
- **What It Means:** Delivers a crucial negative engineering result: under 90–99% packet loss and multi-second latency, radio-based confirmation handshakes (`commit`) and stale beacon guards (`state_guard`) are *not* what prevent physical collisions—local lidar stopping margins and randomized $U[6, 12]$ s park-release timers carry safety even when all three radio guards are turned off. Worse, enabling `commit` under extreme loss adds $+3.1$–3.8 s delay and *increases* transient dual-yield starts (15–16% $\rightarrow$ 22–24%) by holding robots undecided until the 3.0 s timeout triggers simultaneous reactive fallbacks.

3.4 Table IV: 496-Run Seed-Paired ROS 2 / Gazebo Harmonic Transfer Summary

File / Label: `Table IV (label: tab:gz)` | **Manuscript Location:** Section IX-B (RQ4), Page 11
Generator / Source: Authored in `papers/paper1_journal/main.tex` (lines 387–411)
Primary Data Inputs: `data/tables/gazebo/gazebo_all.md`, `gazebo_main_runs.csv`, `gazebo_all_runs.csv`, and `reports/gazebo/hold_for_map_results.md`

- **What It Shows:** A 16-row × 6-column breakdown of all 460 main Gazebo validation runs (14 scenario/study groups covering `S1`, `reactive`, forced yielders, `S2`, `S4`, `S7`, `S9`, `S3`, `S8b/c`, Convoy, T-junction, Asymmetric maps, Blocked bays, and Explore) plus the 36 Batch-2 runs (`hold_for_map` and `det_3.5` rerun). Columns report *Runs*, *Successes*, *Collisions*, *Same First Yielder count* vs. paired 2D seed, and *Mean Delay Offset* $\Delta D = D_{\mathrm{Gz}} - D_{\mathrm{2D}}$ *[s]*.
- **What It Means:** Quantifies exact sim-to-3D transfer fidelity across the entire benchmark:
  1. *Zero-collision safety:* $486/486$ non-reactive Gazebo episodes succeeded with 0 collisions (0.85% 95% upper bound on the 450 main runs), whereas uncoordinated `reactive` suffered 1 collision in 10 Gazebo runs.
  2. *Sub-second straight-corridor fidelity:* $393/406$ (96.8%) negotiating episodes chose the exact same yielder as 2D, and deterministic straight/convoy groups exhibit $|\Delta D| \leq 1.2$ s.
  3. *Pinpointed 3D divergences:* Large $|\Delta D|$ values isolate purely to T-junction stem-release races ($\Delta D = -16.8$ s) and `hold_for_map` lidar-clustering delay ($\Delta D = +10.0$ s), telling practitioners exactly which multi-robot/sensor interactions require 3D testing.

3.5 Table V: Practitioner Policy and Protocol Selection Matrix

File / Label: `Table V (label: tab:guide)` | **Manuscript Location:** Section X-A (Discussion), Page 12
Generator / Source: Authored in `papers/paper1_journal/main.tex` (lines 413–425)
Primary Data Inputs: Distills quantitative decision rules across RQ1–RQ4 (> 50,000 2D + 496 Gazebo episodes)

- **What It Shows:** A 5-row × 4-column engineering decision table mapping five concrete warehouse deployment regimes—(1) homogeneous cooperative fleet with uncertain kinematics ($\sigma \geq 0.5$); (2) heterogeneous urgent, loaded, and mixed-speed cooperative fleet ($\sigma \leq 0.25$); (3) self-interested / multi-vendor shared aisles; (4) multi-robot convoys and T-junction branches; and (5) legacy bay-less aisles with off-lane staging—to *Recommended Policy*, *Protocol Configuration Flags*, and *Empirical Basis & Caveats*.
- **What It Means:** Translates the paper's 18 experimental studies into immediately actionable architecture guidelines: rather than claiming a single policy wins everywhere, it specifies exact quantitative thresholds ($\sigma = 0.5$ noise crossover, $\lambda = 0.25$ fairness credit, unconditional `share_map` over `share_on_detect`, Clarke VCG only when multi-vendor incentives outweigh 29–44% bank surplus, and mandatory `make_room` and `goal_wait` rules for ≥ 3-robot topologies).